

KAGE Fitness Studio ERP — Process Flow Artefacts

A portfolio-ready document showing how manual gym operations can be converted into structured ERP-supported workflows through process analysis, decision points, and business rule alignment.

Project Type

Independent Business Analyst Portfolio Case Study

Domain

Fitness Studio / ERP / Gym Operations

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Date

May 2026

Portfolio Boundary Note This is a fictional BA portfolio artefact inspired by real-world gym operations. It is not a live client project, production ERP implementation, or final technical workflow specification. The purpose is to demonstrate Business Analyst thinking through workflow analysis and process improvement.

DOCUMENT PURPOSE

Purpose of This Artefact

This artefact shows the movement from manual gym operations to ERP-supported workflows. It demonstrates BA skills in process analysis, workflow simplification, business rule alignment, and stakeholder communication.

The flows are written in a simple, recruiter-friendly format that is suitable for PDF conversion and easy portfolio presentation.

PROCESS OVERVIEW

Workflow Summary

Flow Name	Business Area	Current Problem	ERP Improvement	BA Skill Demonstrated
AS-IS vs TO-BE Gym Operations	Overall Operations	Manual records, scattered follow-ups, and limited visibility.	Digital workflows, clearer ownership, and dashboard visibility.	Process mapping and transformation thinking.
PT Client Allocation Flow	PT Session & Trainer Management	Trainer assignment is done informally and can create conflicts.	Structured trainer selection, conflict checks, and package tracking.	Decision-point analysis.
Freelancer Billing Flow	Freelancer Management	Client-based billing is difficult to track manually.	Monthly payable calculation with access-fee based logic.	Business rule alignment.
Renewal & Retention Flow	Membership Management	Expiry follow-up is often manual and delayed.	Renewal queues and expiry tracking improve follow-up control.	Retention workflow design.
Equipment Maintenance Flow	Equipment Maintenance	Repair history is not consistently captured.	Maintenance logs and replacement review flags improve visibility.	Operational control mapping.

Gym Operations Comparison

BUSINESS AREA	AS-IS MANUAL PROCESS	TO-BE ERP-SUPPORTED PROCESS	BUSINESS BENEFIT
Member Registration	Spreadsheets, paper notes, and verbal checks.	Digital member profile with validation and structured member type.	Fewer duplicate records and cleaner data.
PT Session Booking	Trainer and client timings are managed informally.	ERP session logging with package linkage and schedule visibility.	Improved control over session planning.
Trainer Availability	Availability is checked through calls or messages.	Trainer availability view with load and slot awareness.	Reduced scheduling conflicts.
Payments & Invoices	Receipts and payment follow-up are tracked manually.	Invoice tracking with payment status and outstanding balance visibility.	Better financial clarity and follow-up.
Freelancer Billing	Access-fee calculations are done manually.	Monthly billing record based on active client count and own fee.	More consistent billing and fewer errors.
Renewals	Expiry checks are delayed or remembered late.	Renewal queue and expiry alerts support timely follow-up.	Improved retention management.
Equipment Maintenance	Repair notes are informal or incomplete.	Maintenance logs capture issue, cost, vendor, and status.	Traceable repair history and review flags.
Management Reporting	Month-end reporting is prepared from multiple sources.	Dashboard and summary reports provide faster visibility.	Quicker decision-making.

Process transformation value

AS-IS focuses on manual effort, while TO-BE shows how the ERP improves visibility, accountability, and operational consistency.

Trainer Allocation Flow

STEP 01

New PT enquiry

Client expresses interest in personal training.

STEP 02

Capture client goal, package type and preferred timing

Admin records the starting details for allocation.

STEP 03

Check trainer availability, PT load, slots and specialisation

System supports a fit-for-purpose trainer selection.

DECISION 01

Is a suitable trainer available?

Availability, load, timing and specialisation are checked.

YES PATH

Select suitable trainer

Trainer is chosen based on business fit and continuity.

NO PATH

Suggest alternate trainer or time slot

Admin adjusts the allocation and repeats the availability check.

DECISION 02

Is the selected slot conflict-free?

The system checks overlapping sessions for the same trainer.

YES PATH

Create PT session schedule

Final trainer, date and time slot are stored in the ERP.

NO PATH

Block booking or require approved group/shared PT override

If not approved, admin must choose another slot or trainer.

STEP 04

Trainer conducts session

Session proceeds as planned.

DECISION 03

Was the session completed?

Only completed sessions reduce package balance.

YES PATH

Mark Completed and reduce package balance by 1

CompletedSessions increases and RemainingSessions decreases.

NO PATH

Mark Cancelled / No Show / Rescheduled and log reason

Pre-start cancellation does not reduce package balance; reason and timestamp are retained.

STEP 05

Add trainer progress note

Progress tracking supports client continuity.

STEP 06

CommissionStatus = Estimated

Commission is calculated but not yet finalised.

STEP 07

Admin/Finance payout review

Approval is controlled by the payout cycle.

END STATE

CommissionStatus = Approved / Paid

Final commission status is recorded after approval.

Business rule notes

Same trainer should continue with the PT client wherever possible for one package cycle or one month. PT balance reduces only after SessionStatus = Completed. Cancelled sessions before start time do not reduce package balance. Double booking is blocked unless shared/group PT is intentionally approved.

FREELANCER BILLING

Monthly Access-Fee Flow

STEP 01

Register freelancer as MemberType = Freelancer

Freelancer is maintained as a member record.

STEP 02

Link freelancer clients as MemberType = Freelancer Client

Each client is connected to one freelancer.

STEP 03

Count active freelancer clients for billing month

Active client count is captured for that month only.

STEP 04

Apply monthly formula

$\text{MonthlyPayable} = \text{OwnFee} + (\text{ActiveClientCount} \times \text{PerClientAccessFee})$.

STEP 05

Generate monthly billing record

A single bill is created for the month.

DECISION 01

Is payment received?

PaymentStatus is checked after billing is generated.

YES PATH

PaymentStatus = Paid; AccessFlagStatus = Clear

Freelancer access remains clear for the period.

NO PATH

PaymentStatus = Pending / Overdue

The account remains visible for admin follow-up.

DECISION 02

Is overdue threshold crossed?

Configured threshold is checked before gate flag activation.

YES PATH

AccessFlagStatus = Gate Flag Active

Account is flagged for admin review.

NO PATH

Keep status Pending / Warning

Continue normal follow-up until the threshold is crossed or payment is received.

END STATE

Admin reviews or overrides gate flag

Final decision remains with admin.

Business rule notes

Freelancer billing is monthly. Freelancer billing is access-fee based, not session-based. Private PT/coaching income collected by the freelancer is outside KAGE billing scope.
Freelancer client count may change monthly.

RENEWALS & RETENTION

Renewal Follow-Up Flow

STEP 01

System checks membership expiry date

Expiry monitoring runs as part of routine membership tracking.

DECISION 01

Is expiry within 7 days?

The system checks whether the member is entering the renewal window.

YES PATH

Show Renewal Due flag and add to follow-up queue

Front desk/admin can see who needs renewal action.

NO PATH

No renewal action yet; continue monitoring

Member remains in normal active status until the renewal window begins.

STEP 02

Front desk/admin follows up manually

Manual call, note, or in-person follow-up is accepted in MVP.

DECISION 02

Has member renewed?

Payment and renewal status decide the next path.

YES PATH

Record payment, extend membership and remove from queue

Member status stays Active or becomes Renewed as per business terminology.

NO PATH

Expiry date passes without renewal

Member status becomes Expired and remains visible for follow-up.

DECISION 03

Has grace period passed?

Only expired members beyond the grace period move to overdue follow-up.

YES PATH

Move to Overdue Renewal queue

Front desk/admin prioritises follow-up or marks inactive based on policy.

NO PATH

Keep in normal follow-up queue

Continue manual follow-up until renewal, grace expiry, or admin decision.

END STATE

Renewed, Overdue, or Inactive based on business decision

The ERP makes the follow-up state visible, but staff/admin owns the action.

Business rule notes

WhatsApp/SMS automation is outside MVP scope. Manual call, note, or follow-up state is accepted in MVP. Expired members should be visible for follow-up. Access restriction after expiry can be handled by admin/front desk policy.

EQUIPMENT MAINTENANCE

Repair and Review Flow

STEP 01

Equipment issue reported

A problem is observed by staff, trainer, or admin.

STEP 02

Staff/Admin logs maintenance issue

A maintenance entry is created and linked to one equipment record.

STEP 03

Add issue notes, repair date, vendor, and cost

Repair details are captured for traceability and reporting.

STEP 04

CurrentStatus = Under Repair

The equipment status is updated while repair is in progress.

DECISION 01

Is repair completed?

Resolved and unresolved outcomes are separated.

YES PATH

ResolutionStatus = Resolved; CurrentStatus = Operational

The machine returns to service.

NO PATH

ResolutionStatus = Pending Vendor / Escalated

CurrentStatus remains Under Repair or Out of Service until resolved.

STEP 05

System checks repair history

Repeated repair frequency is reviewed over a defined period.

DECISION 02

Has repeated repair threshold been crossed?

The repair count and period are checked against the review rule.

YES PATH

ReplacementReviewFlag = Yes

Owner/Admin is prompted to review repair vs replacement.

NO PATH

Keep monitoring equipment

No replacement review is needed yet; repair history remains recorded.

END STATE

Owner/Admin decides repair, monitor, or replace

The system flags review only; final decision remains with the business.

Business rule notes

System flags replacement review only. Final replacement decision is made by owner/admin. RepairCost must be zero or greater. Each maintenance log must be linked to one equipment item.

Process Flow to Business Rules

PROCESS FLOW	KEY BUSINESS RULES SUPPORTED	RELATED ARTEFACTS
PT Client Allocation	Trainer availability, double-booking control, package balance deduction, session completion logic.	Business Rules Summary, Data Dictionary, UAT Summary
Freelancer Billing	Monthly payable formula, monthly billing cycle, gate flag logic, access-fee model.	Business Rules Summary, Logical Data Model, Data Dictionary
Renewal & Retention	Renewal Due flag, overdue renewal queue, manual follow-up, expiry status update.	Business Rules Summary, UAT Summary
Equipment Maintenance	Maintenance log linkage, repair cost validation, replacement review flag.	Data Dictionary, Business Rules Summary
AS-IS vs TO-BE	Overall workflow transformation from manual tracking to ERP-supported visibility.	ERP Overview Document